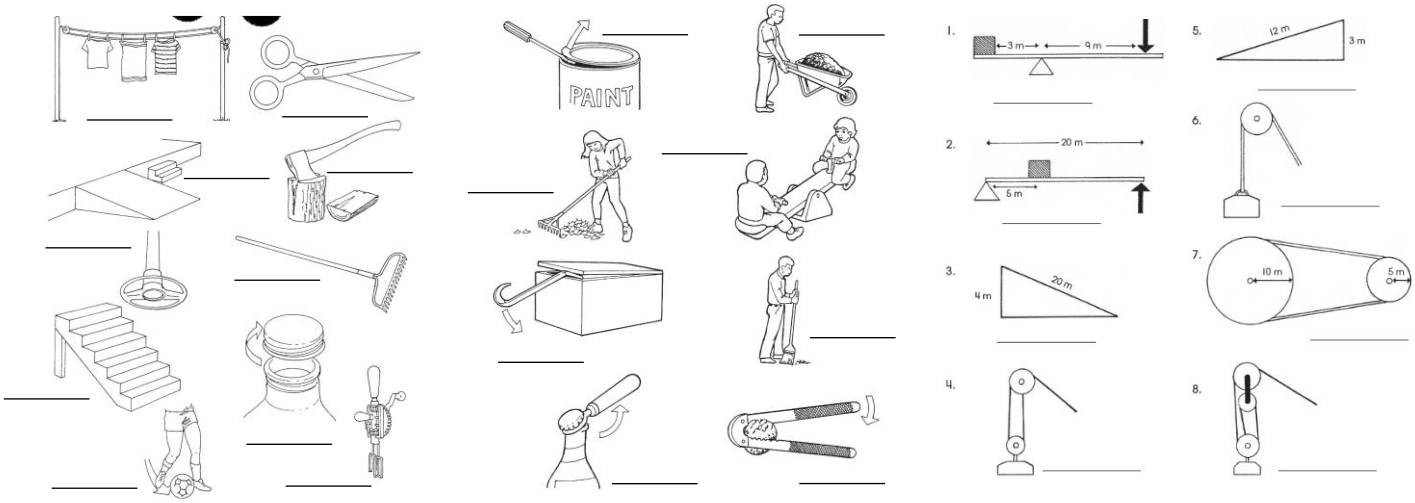
Chapter2 Simple machine Lesson1 (Levers)

I. Levers (section1)

1. Identify the type of simple machine and mechanical advantage as shown in each picture.



2. Explain how a lever can act as a force multiplier.

3. Construction worker uses a board and log as a lever to lift a heavy rock. If the input arm is 3 meters long and the output arm is 0.75 meters long, what is the mechanical advantage of the lever?

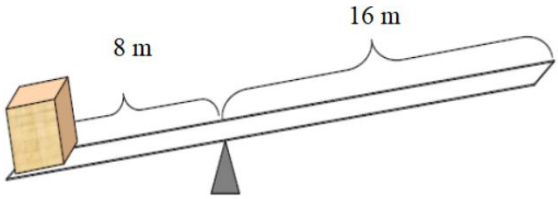
4. A force of 30 newtons is applied to a machine through a distance of 10 meters. The machine is designed to lift an object to a height of 2 meters. If the total work output for the machine is 18 newton-meters (N-m), what is the mechanical advantage of the machine?

5. The mechanical advantage of a lever is 6. How much **force** needs to be exerted on the lever to lift an object with a gravitational force of 320 N?

6. A broom with an input arm length of 0.4 meters has a mechanical advantage of 0.5. What is the length of the output arm?

7. A child's toy rake is held so that its output arm is 0.75 meters. If the mechanical advantage is 0.33, what is the input arm length?

8. Calculate the mechanical advantage of this lever.



9. Using a lever, a person applies 80N of force and moves the lever 1 meter. This moves a 200 N rock at the other end by 0.2 meters. What is the efficiency of the machines?

10. A machine has a mechanical advantage of 5. If 300 N of input force is used to produce 3,000 Nm of work,

a. What is the output force?

b. What is the distance over which the work is applied?

11. An input force of 50 N is applied through a distance of 10 meters to a machine with a mechanical advantage of 3. If the work output for the machine is 450 Nm and this work is applied through a distance of 3 meters, what is the output force of the machine?

12. The mechanical advantage of a lever is 12. How much force needs to be exerted on the lever to lift an object with a gravitational force of 220 N?

13. A child's toy rake is held so that its output arm is 0.75 meters. If the mechanical advantage is 0.84, what is the input arm length?

14. Using a lever, a person applies 90 N of force and moves the lever 3 meter. This moves a 600 N rock at the other end by 0.6 meters. What is the efficiency of the machines?

Chapter2 Simple Machine Lesson (Levers)

II. Lever (section2)

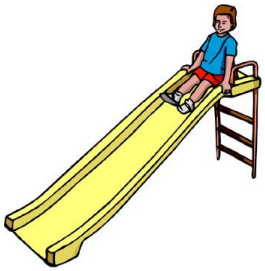
1. What type of simple machine is found on a water bottle cap?
a. lever **b.** pulley **c.** wheel and axle **d.** screw

2. Which is an example of a wheel and axle?
a. shovel **b.** water faucet knob **c.** seesaw **d.** crow bar

3. What type of simple machine is shown in the picture to the right?
a. inclined plane **b.** pulley **c.** wheel and axle **d.** wedge

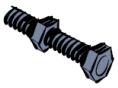
4. To pry the lid off a paint can, you apply a force of 50 N to the handle of the screwdriver. What is the mechanical advantage of the screwdriver if it applies a force of 500 N to the lid?

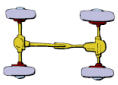
5. On the line by each picture, write the type of simple machine.






and




6. To open a bottle, you apply a force of 50 N to the bottle opener. The bottle opener applies of a force 775 N to the bottle cap. What is the mechanical advantage of the bottle opener?

7. A machine uses an input force of 200 newtons to produce an output force of 800 newtons. What is the mechanical advantage of this machine?

8. Another machine uses an input force of 200 newtons to produce an output force of 80 newtons. What is the mechanical advantage of this machine?

9. A machine is required to produce an output force of 600 newtons. If the machine has a mechanical advantage of 6, what input force must be applied to the machine?

10. A machine with a mechanical advantage of 10 is used to produce an output force of 250 newtons. What input force is applied to this machine?

11. A machine with a mechanical advantage of 2.5 requires an input force of 120 newtons. What output force is produced by this machine?

12. An input force of 35 newtons is applied to a machine with a mechanical advantage of 0.75. What is the size of the load this machine could lift (how large is the output force)?

13. A machine is designed to lift an object with a weight of 12 newtons. If the input force for the machine is set at 4 newtons, what is the mechanical advantage of the machine?

14. A lever has an effort arm with a length of 2 m and a resistance arm with a length of 0.5 m. When 900 N of force is applied to the lever, a 1750-N load is raised 0.3 m.

Chapter2 Simple Machines Lesson 2 (Inclined Plane)

II. Inclined Plane

1. A ramp with a mechanical advantage of 6 is used to move a 36-newton load. What input force is needed to push the load up the ramp?

2. A 10 meters ramp lifts objects to a height of 0.86 meters. What is the mechanical advantage of the ramp?

3. A child makes a ramp to push his toy dump truck up to his sandbox. If he uses 50 Newton of force to push the 140 newtons truck up the ramp, what is the mechanical advantage of his ramp?

4. It takes 200 N to pull an object up an inclined plane. The gravitational force on the object is 800 N.

a. What is the load force in this case?

b. What is the effort force in this case?

c. Calculate the Mechanical Advantage (MA)

5. A 15-meter ramp lifts objects to a height of 5 meters. What is the mechanical advantage of the ramp?

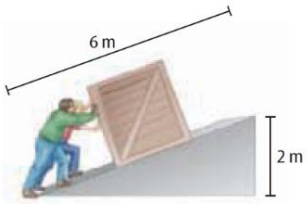
6. A 20-meter-long ramp has a mechanical advantage of 5. What is the height of the ramp?

7. A ramp with a mechanical advantage of 10 lifts objects to a height of 4.5 meters. How long is the ramp?

8. A child makes a ramp to push his toy dump truck up to his sandbox. If he uses 15 Newton of force to push the 25-newton truck up the ramp, what is the mechanical advantage of his ramp?

9. Gina wheels her wheelchair up a ramp using a force of 80 Newton. If the ramp has a mechanical advantage of 7, what is the output force (in Newton)?

10. Using a ramp 6 m long, workers apply an input force of 1250 N to move a 2000-N crate onto a platform 2 m high. What is the efficiency of the ram



11. A slope of length 20 m that rises to a height of 5 m. Determine the mechanical advantage.

12. A slope of length 20 m that rises to a height of 5 m.

13. A slope of length 100 m that rises to the same 5 m height.

14. A slope of length 100 m that rises to the same height. Calculate the mechanical advantage.

15. A slope that is at an angle of 30° to the horizontal and rises to a height of 50 m. Calculate the mechanical advantage.

16. A box weighing 50 kg is placed on a frictionless inclined plane with an angle of inclination of 30 degrees. Calculate the work done in moving the box up the incline by a distance of 10 meters.

17. A person pushes a crate weighing 100 N up an inclined plane that is 5 meters long and inclined at an angle of 45 degrees. If the person applies a force of 150 N parallel to the incline, calculate the work done by the person.

18. A block 40 kg slides down a frictionless inclined plane with an angle of inclination of 60 degrees. If the height of the inclined plane is 10 meters, calculate the work done by gravity on the block.

19. A person pushes a 20 kg block up a frictionless inclined plane that is 4 meters long and inclined at an angle of 30 degrees. If the person applies a force of 100 N parallel to the incline, calculate the work done by the person.

20. A box weighing 80 N is pushed up a frictionless inclined plane with an angle of inclination of 45 degrees. If the box is pushed a distance of 6 meters, calculate the work done in moving the box.

21. A person pushes a 30 kg crate up an inclined plane that is 6 meters long and inclined at an angle of 45 degrees. If the person applies a force of 200 N parallel to the incline, but there is a frictional force of 50 N opposing the motion, calculate the net work done on the crate.

22. A box weighing 80 kg is pushed up a frictionless inclined plane with an angle of inclination of 30 degrees. If the box is pushed a distance of 12 meters, calculate the work done in moving the box.

Chapter2 Simple machine Lesson3 (Pulley)

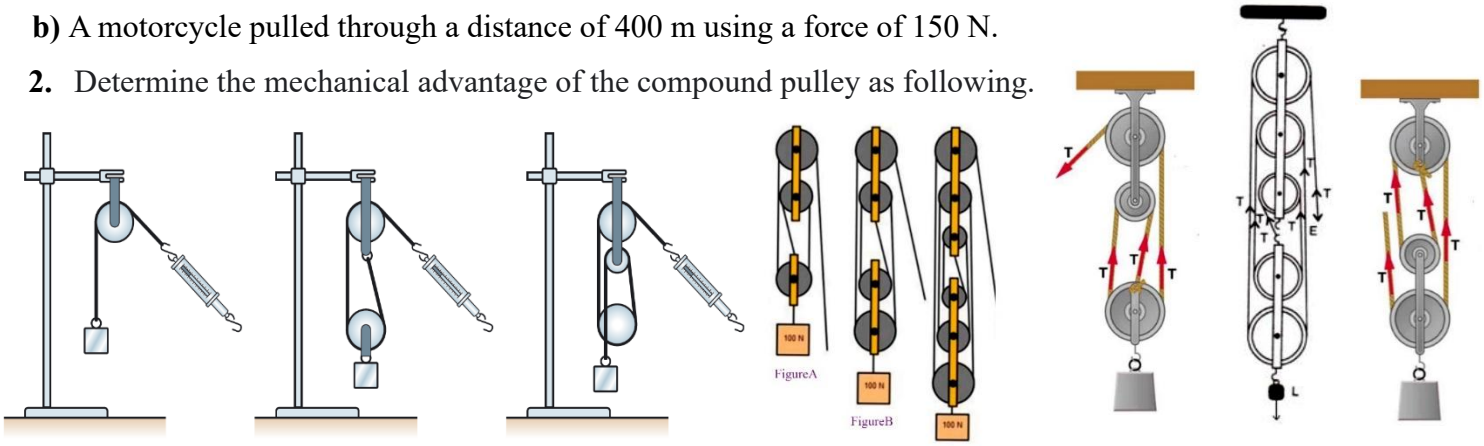
III. Pulley

1. Which of these transfers the most energy? Show your working out.

a) A car being pushed through 100 m using a force of 700 N.

b) A motorcycle pulled through a distance of 400 m using a force of 150 N.

2. Determine the mechanical advantage of the compound pulley as following.


100 N
FigureA
100 N
FigureB
100 N
T
T
T

15. A single fixed pulley is used to lift a load of 800 N. If the efficiency of the pulley system is 85%, calculate the effort required.
16. A single fixed pulley has a mechanical advantage of 3:5. Calculate the effort required to lift a load of 150 kg.
17. A movable pulley system consists of two pulleys. If the load is 200 N, calculate the effort required if the efficiency of the system is 80%.
18. A movable pulley system consists of two pulleys. The mechanical advantage is 2.5. Calculate the effort required to lift a load of 120 N.

Chapter2 Simple machine Lesson 4 (Gears and belt system)

IV. Gear and Belt System

1. A 36-tooth gear turns three times. It is connected to a 12-tooth gear. How many times does the 12-tooth gear turn?
2. A 12-tooth gear is turned two times. How many times will the 24-tooth gear to which it is connected turn?
3. A 60-tooth gear is connected to a 24-tooth gear. If the smaller gear turns ten times, how many turns does the larger gear makes?
4. A 60-tooth gear is connected to a 72-tooth gear. If the smaller gear turns twelve times, how many turns does the larger gear makes?
5. A 72-tooth gear is connected to a 12-tooth gear. If the large gear makes one complete turn, how many turns does the small gear make?
6. A 72-tooth gear is connected to a 12-tooth gear. If the large gear makes 3 complete turns, how many turns does the small gear make?
7. A 60-tooth gear is connected to a 72-tooth gear. If the smaller gear turns 18 times, how many turns does the larger gear make?
8. A 60-tooth gear is connected to another gear. When the 60-tooth gear completed 4 revolutions, the second gear completed 6 revolutions. What will be the number of teeth of second gear?
9. A two-gear system has one wheel with 50 teeth and the other wheel with 5 teeth. If the smaller gear rotates 15 times, how many times does the larger turns? Find the gear ratio of a gear system.
10. A gear system has the driving wheel of radius 2 cm and driven wheel of radius 20 cm.
a) Find the gear ratio
b) If the number of rotations made per minute by the driving wheel is 100, find the number of rotations per minute made by the driven wheel.
c) If the driven wheel has 40 teeth, find the number of teeth in the driving wheel.
11. A gear system has one wheel with 10 teeth and the other wheel with 50 teeth. Calculate the gear ratio.
12. The driving wheel of a gear system is of radius 16 cm and it has 32 teeth. For one rotation of the driving wheel, if the driven wheel makes four rotations, find: (a) the radius, and (b) the number of teeth in the driven wheel.
13. What is the gear ratio between gears 1 and 2 if: Gear 1 has ____A___ teeth and gear 2 has ____B___ teeth.

RATIO	HOW MANY TIMES
A= 10; B = 20 _____	If A turns one time, B will turn _____.
A= 10; B = 40 _____	If A turns two times, B will turn _____.
A= 3; B = 27 _____	If B turns one time, A will turn _____.
A= 10; B = 200 _____	If B turns one time, A will turn _____.
A= 12; B = 48 _____	If A turns one time, B will turn _____.
A= 26; B =130 _____	If A turns five times, B will turn _____.
A= 5; B = 55 _____	If B turns one time, A will turn _____.

14. If the first gear has ____A___ teeth and gear 2 has ____B___ teeth, and gear 3 has ____C_____ teeth

	RATIO HOW	MANY TIMES
A= 10; B = 20; C = 10	**1:20:1_=_1:1**	If A turns one time, C will turn **1**
A= 10; B = 40; C = 5	_____	If A turns one time, C will turn _____
A= 3; B = 27; C = 3	_____	If C turns one time, A will turn _____
A= 10; B = 200; C = 200	_____	If B turns one time, C will turn _____
A= 12; B = 48; C = 6	_____	If A turns one time, C will turn _____

15. In figure 5.34 the teeth of gear wheels exert a force of 250 N on each other. The radius of the smaller wheel is 10 cm, and the larger wheel has a radius of 20 cm.

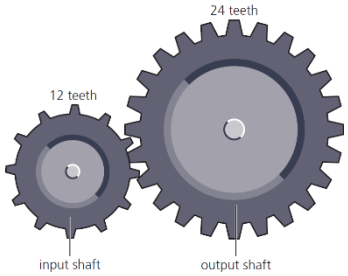
a) Calculate the turning moment produced by each wheel.

b) Explain how gears can be used in machines.

c) Which gear wheel is likely to wear out first?

d) If the driver gear rotates 4 times, how many turns does the driven gear turns?

e) What is the gear ratio?



▲ Figure 5.34

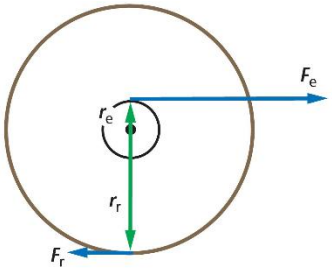
16. You are studying the rear wheel on your bicycle. It has a radius of 35.6 cm and has a gear with radius of 4.00 cm. When the chain is pulled with a force of 155 N, the wheel rim moves 14.0 cm. The efficiency of this part of the bicycle is 95%.

a) What is the IMA of the wheel and gear?

b) What is the MA of the wheel and gear?

c) What is the resistance of force?

d) How far was the chain pulled to move the rim that amount?



Hint:

$$MA = \frac{F_R}{F_E} \qquad eff = \frac{W_O}{W_{in}} \times 100\%$$

$$IMA = \frac{d_E}{d_R} \qquad eff = \frac{MA}{IMA} \times 100\%$$

17. A large gear with 48 teeth is connected to a small gear with 12 teeth. If the large gear turns twice, how many times the small gear turn?

18. A gear system consists of two gears. The first gear has a radius of 10 cm and the second gear has a radius of 5 cm. If a force of 20 N is applied to the first gear, what is the output force exerted by the second gear?

19. A bicycle has a gear system with a chaining of 40 teeth and a rear cog with 20 teeth. If a force of 100 N is applied to the pedals, what is the output force exerted by the rear wheel?

20. A machine uses a gear system to lift a load. The input gear has a radius of 8 cm and the output gear has a radius of 16 cm. If an input force of 50 N is applied, what is the output force exerted on the load?

21. A gear system is used to lift a weight of 500 kg. The input gear has a radius of 20 cm and the output gear has radius of 40 cm. If an input force of 100 N is applied, what is the mechanical advantage of the gear system?

22. A machine uses a gear system to move a load horizontally. The input gear has a radius of 12 cm and the output gear has a radius of 6 cm. If an input force of 80 N is applied, what is the output force exerted on the load?

23. A gear system is used to lift a load. The input gear has a radius of 15 cm and the output gear has a radius of 5 cm. If an input force of 50 N is applied, what is the mechanical advantage of the gear system?

24. A bicycle has front gear with 40 teeth and a rear gear with 20 teeth. If the cyclist pedals 50 times, calculate the number of turns made by the rear gear.

25. Two gears are meshed together. The larger gear has 40 teeth and make 10 turns. Calculate the number of turns and teeth of the smaller gear if the gears are of the same size.

26. A gear with 30 teeth rotates 5 times. Calculate the number of teeth on a meshed gear if the gears are the same size.

27. A machine has a gear train consisting of three gears. The first has 20 teeth and makes 15 turns. The second gear has 30 teeth. Calculate the number of turns and teeth of the third gear if the gears are the same size.
28. A bicycle has a front gear with 36 teeth and a rear gear with 18 teeth. If the cyclist pedals 80 times, calculate the number of turns made by the rear gear.
29. Two gears are meshed together. The larger gear has a radius of 10 cm and makes 60 revolutions per minute. The smaller gear has a radius of 5 cm. Calculate the number of turns of the smaller gear and gear ratio of a gear system.
30. A conveyor belt system has a driver pulley with a radius of 20 cm and a driven pulley with a radius of 10 cm. If the driver pulley is rotating at 100 revolutions per minute, calculate the number of turns of driven pulley

Chapter2 Simple machine Lesson 5 (Wheel and Axle)

V. Wheel and Axle

1. A wheel of radius $40\ \text{cm}$ is attached to an axle of radius $10\ \text{cm}$. If an effort of $20\ \text{N}$ is applied on the wheel, what is the load lifted?
2. A wheel of $50\ \text{cm}$ radius is attached to an axle of $5\ \text{cm}$ radius. If $100\ \text{N}$ effort is applied, calculate the load lifted.
3. If the wheel radius $= 30\ \text{cm}$ and axle radius $= 10\ \text{cm}$, what is the mechanical advantage?
4. A wheel and axle has a wheel radius of $25\ \text{cm}$ and axle radius of $5\ \text{cm}$. Efficiency $= 80\%$. If the effort is $40\ \text{N}$, what is the load?
5. A wheel of $20\ \text{cm}$ radius and an axle of $4\ \text{cm}$ radius are used to lift a load. If the effort is $25\ \text{N}$, find the load (ignoring friction).
6. If a wheel and axle has a velocity ratio of 5 and efficiency of 80%, its mechanical advantage is:
7. A wheel of radius $50\ \text{cm}$ and axle of radius 10 cm are used. If a load of $200\ \text{N}$ is lifted, what effort is needed (ignoring friction)?
8. In a wheel and axle, the effort arm is $40\ \text{cm}$ and the load arm is $10\ \text{cm}$. If the load is $120\ \text{N}$, find the effort (ideal).
9. A wheel and axle has a wheel radius of $60\ \text{cm}$ and axle radius of $15\ \text{cm}$. If effort $= 25\ \text{N}$, load = ?
10. A wheel of radius $45\ \text{cm}$ and axle radius of $15\ \text{cm}$ is used. If the load lifted is $180\ \text{N}$, find the effort.
11. If the axle radius is increased while keeping wheel radius same, efficiency:
12. A wheel radius $25\ \text{cm}$, axle radius $5\ \text{cm}$. Load $= 200\ \text{N}$. Find effort (ideal).
13. A wheel radius $= 20\ \text{cm}$, axle radius = 10 cm. If effort = 50 N, find load (ideal).
14. If a wheel radius $= 15\ \text{cm}$, axle radius $= 5\ \text{cm}$, effort $= 40\ \text{N}$, efficiency $= 75\%$. Find load.